

COMPUTATION AND VERIFICATION REPORT: REVISION R40

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This report distinguishes inherited experiments, fresh read-only verification, new exact property tests, manuscript generation, and preservation of the complete scientific record. It specifies the actual published objects and executable commands rather than relying on a successful document build as evidence of scientific validity.

1. FROZEN SOURCE AND REVIEW

The addressed report is `reviews/2026-09-25-econometrica-r38/referee_report.md`, at review commit `acbede22e517a245fcc5244de7171c1ae218ea73`. It evaluates the R38 article. The intervening scientific branch already contained the common-continuation solvers, proofs, and checked result objects. Its frozen head is `bf5d42b08debd4c26341ab1b1231a6a6242df4f9`. R40 starts from that head on a new branch:

`revision/econometrica-r40-referee-complete-2026-09-25`.

The new branch preserves both earlier manuscripts and computational objects. Its additions are the witness-driven repair theorem and constructor, the fixed-space model-uncertainty proposition, break-even randomization administration charges, further exact tests, a finer nonlinear independent replay, missing deterministic diagnostics and publication checks, and four complete compiled review documents. R39 computations are not represented as newly performed experiments or a second prospective holdout.

The eight finite-state environments were fixed in the inherited protocol at commit `13fd3ee0eec78db5fa6aed34e63819c5ed546209`. The protocol records development seed 390001 and evaluation seeds 391101–391108, model parameters, both solver hashes, a 4,096-node allowance, a 60-second per-case/per-solver cap, and absolute stopping allowance 10^{-3} . R40 checks the frozen solver hashes and does not tune or rerun the timed searches to select more favorable endpoints. Each published finite row belongs to its retained full proof tree. The 42 primary configurations remain a development cohort.

2. FOUR DIFFERENT VERIFICATION STATEMENTS

2.0.0.1. *Inherited independent arithmetic.* All 16 finite trees and 42 primary lottery/local-bound pairs have retained independent SymPy reports. Their proof-object hashes and report identities are checked before table generation. The primary operating checks cover open cells and isolated points; integrated cost bounds use the explicitly stated absolutely continuous initial laws and nonsingular affine transitions. Such an integral bound is not relabeled as a point-mass guarantee. Historical R34 randomized and R38 deterministic components retain their own provenance when they dominate a displayed endpoint.

2.0.0.2. *Fresh R40 read-only replay.* The R40 wrapper calls the frozen independent verifiers without modifying inherited results. It rechecks all 16 finite trees, the complete primary objects at $(T, \text{rule}, \varepsilon) = (4, \text{neural31001}, 1/100), (8, \text{neural31002}, 1/20), (12, \text{defer}, 1/100)$, and the entire nonlinear $(T, N, \varepsilon) = (32, 128, 1/2)$ object. The nonlinear check reconstructs rational corner geometry, every support/action inequality, the connected lower recursion, and the selected policy's upper cost and all-restart operating bound with scalar arbitrary-precision arithmetic.

This extends the earlier independent nonlinear coverage, which named only (8, 64) and (32, 64). The finest $N = 256$ row is not assigned the independent status of $N = 128$.

Fresh execution records are `results/replay_finite.json`, `results/replay_continuum.json`, and `results/replay_nonlinear.json` under the R40 directory. They include actual runtime, verifier identity, proof identity, and explicit coverage. The three-primary-object fresh replay is not described as 42 new executions. The displayed scientific construction times remain those of the frozen experiments, not times from the read-only replay.

2.0.0.3. *New exact properties.* Seed 400925 generates 48 witness-repair test models with signed rewards, multiple action and state counts, and horizons 2–5. The constructor uses only the proposed policy, primitive model, and operating upper witness. A separate exact operating solve constructs test witnesses and serves as an oracle; it is not passed to the repair function. Fraction calculations are compared against independently assembled SymPy transition matrices and value recursions. All 48 nonpositive-slack inputs are rejected as uncertified. The test does not infer true economic infeasibility from this rejection.

The same audit generates 200 discontinuous piecewise-affine pairs, checks 5,756 exact values of lower envelopes and positive affine pullbacks, and rejects 400 malformed partitions. A further 120 local programs are checked by an independent active-basis matrix calculation. There are 59 cases with coincident operating values and 64 with a pure action on the boundary. These are measured counts from these tests, not retroactive degeneracy counts for the historical adaptive LP. All tests supplement analytic arguments; they do not establish a statistical false-acceptance probability or all-input correctness.

2.0.0.4. *Publication integrity.* The publication audit compares every primary displayed endpoint with its actual components, recomputes all deterministic weighted integrals used in strict separation, checks each finite summary against its full proof tree, and verifies identities of all nonlinear objects and both frozen solvers. Economic overhead margins are computed exactly for all 126 primary initial-law configurations. Tables round lower endpoints downward and upper endpoints upward. Hash equality establishes identity, not a mathematical inequality. The publication manifest records these two kinds of evidence separately.

3. REPRODUCTION

From the repository root, the retained scientific environment uses Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, and SymPy 1.14.0. The inherited CI folders retain wheel identities, package lists, source commits, CPU and operating-system records. R40’s actual environment and package list are stored with its build. These records are an executable recipe, not a claim that a changing hosted runner is an immutable container image. Set `OPENBLAS_NUM_THREADS`, `OMP_NUM_THREADS`, and `MKL_NUM_THREADS` to one, and `PYTHONHASHSEED` to zero. Python optimization must be disabled because assertions are scientific checks.

For read-only R40 verification and publication:

```
r=revisions/2026-09-25-r40
python $r/replication/property_tests.py
python $r/replication/audit_replay.py finite
python $r/replication/audit_replay.py continuum
python $r/replication/audit_replay.py nonlinear
python $r/replication/prepare_revision.py
python $r/replication/make_figures.py
for p in ECTA SUPP RESPONSE COMPUTATION; do
```

```

latexmk -pdf -interaction=nonstopmode \
-halt-on-error ${p}_R40.tex
done

```

```
python $r/replication/prepare_revision.py --manifest
```

The final manifest gate requires completed and successful named replay reports, all four PDFs, clean cross-references, and exact preservation of every inherited tracked blob. A fresh clone can rerun these steps without fitting a new policy. The publication workflow may consume the completed read-only replay reports after checking their exact verifier and object hashes; that consumption is not another independent arithmetic run.

Full scientific reconstruction instead uses `revisions/2026-09-25-r39/replication/run_finite.py`, `randomized_primary.py`, `primary_restart.py`, and `nonlinear_support.py`, followed by their respective independent verifiers. Those commands intentionally generate new proof objects and may overwrite local R39 result files in the working checkout. Run full reconstruction in a separate worktree to keep the frozen publication intact. Timed searches can visit a different number of nodes on a different machine; this is not a failure of exact verification. Neither tables nor manuscripts select the best of multiple timed runs.

4. ERROR, WORK, AND CLAIMS

The witness-driven theorem exposes the sufficient margin $(1 - \beta)\varepsilon - d$. Its complete lattice bracket separates witness width, operating residual, policy resolution, and the repair cost. It does not assert that every arbitrary witness has a positive margin, or that exhaustive search is inexpensive. The continuous-state transfer additionally requires a checked reset-kernel approximation in operator total variation. Atomic-shock weak convergence is not substituted for that assumption.

The finite comparison reports all 16 solver/case rows, including every budget or time stop. The unchanged primary cohort has 27 certified uniform-law deterministic separations, 12 zero-cost optima, and 30 positive randomized intervals with nonzero widths. The local recursion improves 23 uniform lower endpoints. The finest nonlinear horizon-32 interval is $[8.8718, 11.8492]$ and retains relative width 25.13%. Its allowance is $1/2$. The four $1/20$ nonlinear rows remain lower-bound-only and are never combined with upper policies certified only at $1/2$.

The administration-cost calculation is a derived economic robustness margin under the known designed model. It is not calibrated monetary welfare. Positive per-date overhead below its certified threshold preserves a strict deterministic-versus-randomized ranking. Model-uncertainty margins require separately verified primitive error bounds; R40 does not manufacture empirical confidence regions. The stopped-control numerical target and its boundary/action-coverage obligations remain archived and unresolved at the requested allowance, rather than being deleted or treated as solved by a maintenance example.

5. PRESERVATION AND REVIEW NAVIGATION

The current article, supplement, response, and computation report are `ECTA_R40.pdf`, `SUPP_R40.pdf`, `RESPONSE_R40.pdf`, and `COMPUTATION_R40.pdf`. Every source file and generated table is committed beneath the R40 directory. The new supplement appends the entire R38 main article and technical supplement verbatim. The complete `HISTORY_R38.pdf`, old reviews, prior failed runs, and all old sources remain unchanged. The only redirected inherited file is `REVISION_INDEX.md`; its exact previous bytes are retained in the R40 history directory.

The Git-based publication gate compares each tracked blob at the frozen R39 commit with the corresponding working-tree file. It checks all inherited files, not just selected manuscripts.

The resulting count and path-level identities appear in `results/preservation.json`; `PUBLICATION_MANIFEST.json` records the new document identities and scientific inputs. This permits a subsequent referee to distinguish preserved historical evidence from the current manuscript's additional results without losing any earlier technical content.